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THE STUDY OF POLYLACTIC ACID AND NANOCCLAY ON THE DEGRADATION PROPERTIES OF LOW-DENSITY POLYETHYLENE COMPOSITES.I.A. Dina¹, C.E Gimba, A.I. Okele ³.

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ABSTRACT

A study was carried out to check the effect of using polylactic acid, Nano clay and glycerine as compatibilizer on the degradation property of virgin and waste low density Polyethylene which is the major component used in producing polyethylene bag. Virgin and waste samples of polyethylene, polylactic acid and Nano clay were obtained locally and compounded using the two-roll mix and the compression moulding machine to produce the composites. Moulded samples produced were cut accordingly using ASTM specification for the tests. Tests carried out included tensile, flexural, hardness, impact, water absorption, soil burial test. Results showed improved tensile, flexural, hardness strength as compared to the control sample (100% Virgin Polyethylene). The addition of polylactic acid, Nano clay and glycerine showed good interfacial bonding between the matrix and the fillers which improved the mechanical properties, water absorption and aided degradation during soil burial test. As the amount of filler content increased it affected the impact test due to the brittle nature of both Nano clay and polylactic acid.

Keywords: *Low density polyethylene, Nano clay, polylactic acid, glycerine, degradation. physio-mechanical properties.*

INTRODUCTION

Poly(lactide) (PLA) has unique properties like good appearance, high mechanical strength, and low toxicity; and good barrier properties have broadened its applications. Numerous researchers have studied the different properties of PLA alone and in combination with other polymers as blend or copolymer. The studied carried out by Auras and others (2004) on the mechanical, physical, and barrier properties of PLA films. These two films were compared with polystyrene and polyethylene terephthalate (PET). Low glass transition temperature of PLA limits its usages in thermally processed packages. Because of its

deformation and its slow melting temperature, it is better to use it for heat-sealing and thermoforming applications.

PLA films have good tensile strength with higher values than polystyrene but lower than PET had lower T_m (melting point) and T_g (glass transition temperature) than PET which makes PLA better for heat-sealing and thermal processing. In terms of barrier properties of PLA, the permeability coefficients of CO_2 and O_2 were lower than those of PS and comparable to those of PET. For tensile modulus and flexural modulus, PLA has the highest value in comparison to PS, PP, and HDPE (Mbugu, 2017). The disposal problem due to non-

degradable petroleum based plastics has raised the demand for biodegradable polymers as means of reducing the environmental impact. PLA is useful for producing loose-fill packaging, compost bags, food packaging, and disposable tableware. Due to its biocompatibility and biodegradability, PLA has also found ample interest as a polymeric scaffold for drug delivery purposes (Nicolas, 2019).

Polyethylene, with the chemical formula $[-CH_2-CH_2-]_n$ (where n denotes that the chemical formula inside the brackets repeats itself to form the plastic molecule) is made in low- and high-density forms. Low-density polyethylene (LDPE) has a density ranging from 0.91 to 0.93 g/cm³. Polyethylene products are very common in our daily life. For examples, food and pharmaceutical packaging film, wire and cable insulation and pipe. Therefore, the production of polyethylene is humongous as is one of the most popular used polymer materials in day-to-day life. Polyethylene and polypropylene represent about 92% of the synthetic plastics produced, and they are used for production of plastic bags, disposable containers, bottles, packaging materials, etc. (Ploypetchara, 2014). An estimated more than 500 billion to 1 trillion plastic bags used globally disturb the ecosystem and ultimately result into serious environmental issues of recycling these materials from the environment. The waste plastics that form particulate matter by UV irradiation and weathering increase surface area and mobility and thereby incorporate easily into the food chain, causing fatal injuries to all

living organisms. The long-range structure and morphology of polyethylene have shown important roles, with amorphous regions being more prone to microbial attack than crystalline ones. This review focuses on the recent hypotheses and experimental findings regarding the biodegradation of polyethylene. (Li, 2018). The poly(lactic acid) or PLA is promoted as the environmental friendly material. Because of its property as a biodegradable plastics, PLA is designed to be used in applications such as medical devices, agricultural tools as well as packaging materials (Gupta and Kumar, 2007; Sin et al., 2013). The good degradation rates of PLA was found at/or above glass transition temperature ($T_g \sim 55-60^\circ\text{C}$).

The use of nanoclays in food packaging has become of interest due to their importance in improving mechanical properties and barrier properties (Chaudhary, 2020). Further, nanoclay is abundant, cost-effective, and biocompatible with a diverse morphology and multiple chemical compositions, making it more important over other nanomaterials used in the advancement of materials science and technology.

Nanoclays are layered silicates with single layer of 0.7 nm thickness, while double layer is 1 nm thick. The interlayer can be modified using surfactants to improve the plasticity and swelling capacity. Surface properties of nanoclays depend on nanosheet charge, nature of surface atoms, and interlayer exchangeable cations. The surface of nanoclay is negatively charged owing to the substitution of Al^{+3} or

Mg²⁺ ions with silicon (Abulyazied, 2021).

MATERIALS AND METHODS

Virgin Low Density Polyethylene (Albarka Plastic, Kaduna), waste low density polyethylene (water satchet bags) from Kaduna Polytechnic, Kaduna, Polylactic Acid (JHD, China), nano clay (NILEST, Zaria), Glycerine (J. Chemicals limited Kaduna.)

Two roll mill (model: 5183 North Bergen U.S.A), Wenzhouzhiguang Compression Moulding Machine (model: 0557), Hounds Field Tensometer (model: W6466), Izod Impact tester model : 6757 pianezz-terino Italy), Vickers hardness testing machine (Model : MV1-PC), Analytical Digital Weighing Balance.

Samples Preparation

Using the formulation table below, six composites were produced using the two rolls mill machine in counter clockwise motion for a period of 5 minutes at a temperature of 170°C, upon achieving a band and bank formation of the LDPE on the front roll, polylactic acid was introduced and allowed to mix for 3 minutes with effective cross mixing. Nanoclay was added gradually to the band, cross mixed and allowed to mix for 3 minutes, then glycerine (plasticizer) was added onto the band. The composites sheet were labelled accordingly. The composites samples obtained from the mixing were placed in metal molds and then onto the compression molding machine for shaping at temperature of 150°C and pressure of 25 bar for 5mins with a dimension of 120mm x100mm x 3.2mm followed by cutting and preparation for further testing according to the ASTM standards for testing.

Table 1: Formulation and composition of LDPE/PLA/Clay composites

Sam ple	Low Densit y	Polylacti c Acid (g)	Na no	Glyce rin (ml)
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	Polyet hylene (LDPE) (g)		cla y (g)	
A	100	-	-	-
B	90	5	5	5
C	80	5	10	5
D	70	5	15	5
E	60	5	20	5
F	50	5	25	5

Methods

Determination of mechanical properties.

Tensile test

The tensile strength test was carried out using the Monsanto Tensometer (model no: W6466) according to ASTM D-638. A dumbbell shaped samples with gauge dimensions 50 mm x 10 mm x 3.2 mm were subjected to a tensile force and tensile properties such as tensile strength, Strain, modulus for each sample.

Flexural test

The flexural strength test on the blends was carried out in accordance with ASTM D-790. The specimen measuring 100 mm x 25 mm x 3.2mm was placed on a support span horizontally at 80 mm gauge length and a steady load was applied to the centre by the loading nose producing three-point bending until the sample specimen failed. The maximum load (N) and the corresponding deflection (mm) were recorded accordingly as the sample specimen failed. The flexural strength and flexural modulus were calculated

Impact test

The impact test was carried out according to the standard specified ASTM D-156; the specimen was cut to specimen dimension 64 mm x12.7 mm x 3.2mm and 45° notched was inserted at the middle of the test specimens from all the produced blend samples. The impact energy test was carried out using Izod Impact Tester (Resil impactor testing machine). The specimen was clamped vertically (IZOD) on the jaw of the machine and hammer of weight 1500 N was released from an inclined angle 240°. The impact energy for corresponding tested

specimen was taken and recorded. Impact strength was calculated and recorded accordingly.

Hardness test

The hardness tests were carried out in accordance with ASTM D2240 using Vicker Microhardness Tester. The sample was placed on the mounting stage and the indenter was lower until it come in contact with the sample at uniform pressure the reading indicating resistance to indentation displayed on the machine screen and recoded accordingly. This was repeated three (3) times at different positions on the sample and average hardness was taken using equation.

$$\text{Average Hardness} = (1\text{st} + 2\text{nd} + 3\text{rd}) / 3$$

Water Absorption Test

The water absorption test was carried out on the composites. The samples were immersed in a container filled with water and then allowed to stay for 24 hours after which it was removed from the water, a dry towel was use to dry the water and reweighed to note the final mass. This was carried out for 75 days at an interval of one week. The result obtained was been used to calculate the percentage water absorption. Using the equation

$$\% \text{ Water absorption} = (W_f - W_i) / W_i \times 100 / 1$$

Where;

W_f = Final weight of sample

W_i = Initial weight of sample before immersion in water

RESULTS

Tensile results

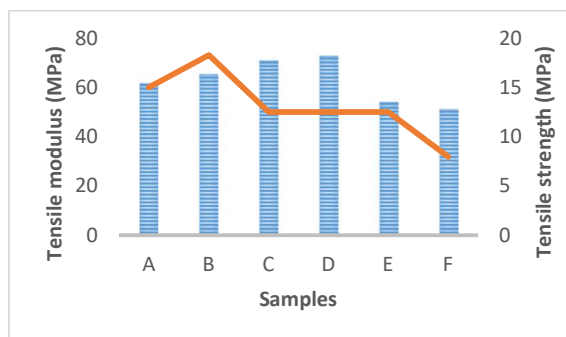


Figure 1: Result of effect of natural fillers on the tensile strength and modulus of polyethylene composite

Figure 1 showed that the control sample A had a tensile strength of 15.38MPa. Sample B had a tensile strength of 16.31 MPa. Sample C has the tensile strength of 17.66 MPa. Sample D has the highest tensile strength of 18.20 MPa. As the amount of filler content increased in sample E and F, a decrease in tensile strength was seen in sample E (13.50 MPa) and sample F has the lowest tensile strength of 12.75 MPa). Sample A tensile modulus was 60.07 MPa. From sample B to F the sample tensile modulus decreased as the amount of filler content increased. Sample F become more brittle as compared to sample B this is because more clay is in the sample hence making it more hard and brittle, the nanosheet surface is hydrophobic owing to Si–O covalent linkages. The hydrophilicity of nanoclays can be achieved using exchangeable hydrophilic cations. Nanoclay particles can absorb water and stacked silicates layers may swell. Incorporation of clay nanoparticles in polymers reveal improved mechanical, thermal, electrical, flame resistant, and gas barrier properties (Calambas, 2021). The introduction of nano clay into bio-based food packaging materials decreases the elongation at break value of the films, and thus overall elasticity of the nano composite is reduced by the inclusion of the nanomaterial. This is because nanoclay's stiffening and reinforcing properties create a network of polymeric phases and reduce the mobility of polymer chains as observed by the studies of (Radfar, 2021).

Flexural test results

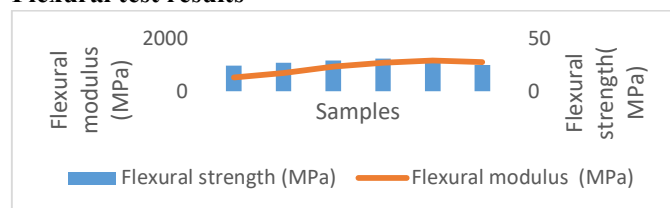


Figure 2: Result of effect of natural fillers on the flexural strength and modulus of polyethylene composite

Figure 2 showed that sample A had a flexural strength of 23.92MPa and a general increase was seen in the flexural behaviour (strength) up to sample D (30.94MPa). Sample E showed a decrease in flexural strength 27.36 MPa

followed by Sample F which had a value of 24.67 MPa. The flexural modulus for sample A was 523.776 MPa and the modulus kept increasing for sample B (692.76), C (929.67), D (1071.99) and E (1156.06). Hence the addition of nanoclay and polylactic acid increased the flexural modulus.

Impact test

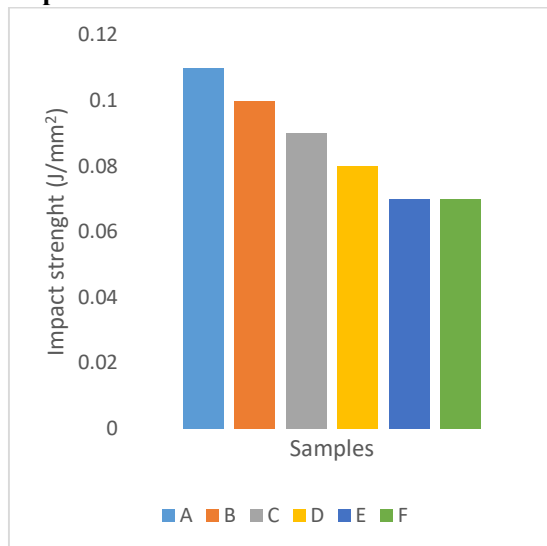


Figure 3 Result of effect of natural fillers on the impact strength of polyethylene composite

Results showed a decrease in impact strength as seen from sample A to sample F. The results revealed that as the amount of nanoclay and polylactic acid content increased, it decreased the impact strength, this could be attributed to the brittle nature of nanoclay and polylactic acid. The presences of LDPE particles between PLA chains also increased the mobility of PLA chains allowing it to deform easily (Radfar, (2020). Although the enhanced mobility of PLA chains might pass a positive effect to PLA/LDPE blends in terms of impact strength properties, but it had led to reduction of rigidity of the blends (Lu, 2005).

Hardness test

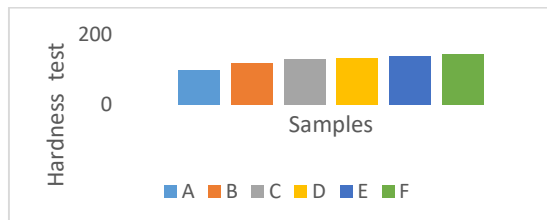


Figure 4: Result of effect of natural fillers on the hardness of polyethylene composite

Results showed an increase in the hardness of the composite. Sample A has the average hardness of 106.3 while sample F has the average hardness of 145.9. This can be attributed to the good interfacial bonding between the PE / nanoclay / Polylactic acid using glycerine as a compatibilizer. The primary purpose of incorporating a nanoparticle (reinforcement agent) into a biopolymer matrix is to improve the mechanical properties of the packaging material (Emadian, 2017).

Water absorption

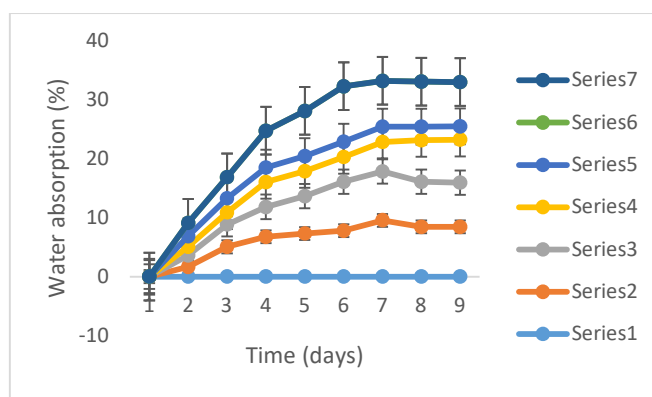


Figure 5: Result of effect of natural fillers on the water absorption properties of polyethylene composite

Results showed that all samples absorbed water every week due to the hydrophilic Polylactic acid and clay. Samples B, C, D showed maximum water absorption up to forty two days (7 week) and a decline in the water absorption rate. Sample E that contains 25% nano clay kept absorbing water even after 75 days as studied by other scholars. The water absorption capacity of nanoclays depends on the number of exchangeable cations in interlayer. PLA a biodegradable and compostable polymer was used to evaluate the biodegradable nature of nanoclay. The results of the biodegradation experiments demonstrate that the

biodegradation phase of nanoclay containing films began earlier than that of pure PLA films (Chadha, 2022). The influence of incorporation of clay nanoparticles in polymers reveal improved mechanical, thermal, electrical, flame resistant, and gas barrier properties (Abulyazied, 2021). The high surface area and nanometric size of the clays enable them to form an efficient interaction for transmitting tensile stress, nanoclays reduce the rate of biodegradation of the packaging materials. Nevertheless, most importantly, when considering the limited studies performed the migration levels of nanoclay are very low while it is non-toxic. A higher concentration of nanoclay film (1.5%) results in the production of aggregates, which degrades the mechanical characteristics.

Conclusion

1. A bio-composite was made using low density polyethylene, polylactic acid, clay as the filler and glycerine as the compatibilizer to aid the improvement of the physio-chemical properties of the composite as well as improve the degradation properties of the composite as observed in the water absorption test. The utilization of bio-nanocomposites based food novel packaging material has become of great interest in the food packaging industry, mainly due to its importance in the increment of biopolymer properties such as mechanical properties and the barrier properties of the packaging materials.
2. The results revealed that there was an increase in the tensile strength from the control sample A to sample D. Sample D had the best results with 70% LDPE, 5% polylactic acid, 15% nanoclay, 5% glycerine in terms of the tensile and flexural strength pointing to the fact that there was good interfacial cohesion between matrix and filler using glycerine as a compatibilizer and this improved mechanical property as seen. Polylactic acid has very good mechanical and thermal properties as compared to polyethylene and thus it improved the mechanical properties.
3. Results of hardness and water absorption showed an increase as natural filler content increased due to the hydrophilic nature of nanoclay and brittle nature of polylactic acid. The impact strength decreased as natural filler content increased. The nanoclay and polylactic acid aided the biodegradation of LDPE and this was seen in the water absorption test that the biodegradation phase of nanoclay began earlier than that of PLA and this lead to a loss of mass.
4. The bio-composite produced can be used in the production of plastic items with good mechanical and hardness properties such as children toys, waste bins and table tops or any plastic material which water absorbency is not of paramount significance. With the great importance of nanoclay in

enhancing the properties of polymers its industrial applications are very limited, still open to further research with the potency to develop a non-toxic, low-cost, environmental friendly food packaging materials with enhanced properties.

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